

Chapter 11 Part 1: Acidity

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement. A choice may be used once, more than once, or not at all in each set.

Refer to the following types of chemical or physical changes.

- (A) Oxidation-reduction reaction
- (B) Brønsted-Lowry acid-base reaction
- (C) Sublimation
- (D) Dehydration
- (E) Precipitation

1. Occurs when aqueous solutions of ammonia and vinegar are mixed

(A) Oxidation-reduction reaction

(B) Brønsted-Lowry acid-base reaction

(C) Sublimation

(D) Dehydration

(E) Precipitation

2. Occurs when solid sodium acetate, $\text{NaC}_2\text{H}_3\text{O}_2(s)$, is added to water

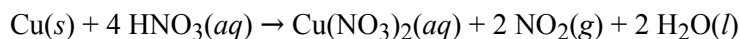
(A) Oxidation-reduction reaction

(B) Brønsted-Lowry acid-base reaction

(C) Sublimation

(D) Dehydration

(E) Precipitation



Each student in a class placed a 2.00 g sample of a mixture of Cu and Al in a beaker and placed the beaker in a fume hood. The students slowly poured 15.0 mL of 15.8 M $\text{HNO}_3(aq)$ into their beakers. The reaction between the copper in the mixture and the $\text{HNO}_3(aq)$ is represented by the equation above. The students observed that a brown gas was released from the beakers and that the solutions turned blue, indicating the formation of $\text{Cu}^{2+}(aq)$. The solutions were then diluted with distilled water to known volumes.

3. Which of the following is true about the reaction?

(A) It is a Brønsted-Lowry acid-base reaction, because the solution is neutral at the end.

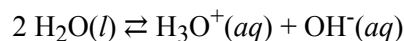
(B) It is a Brønsted-Lowry acid-base reaction, because $\text{HNO}_3(aq)$ is a strong acid.

(C) It is a redox reaction, because $\text{Cu}(s)$ is oxidized and $\text{H}^+(aq)$ is reduced.

(D) It is a redox reaction, because $\text{Cu}(s)$ is oxidized and the nitrogen atom in $\text{NO}_3^-(aq)$ is reduced.

Chapter 11 Part 1: Acidity

4.



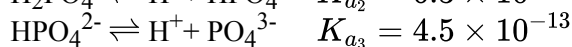
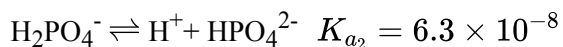
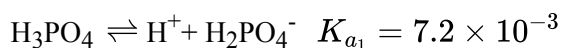
The autoionization of water is represented by the equation above. Values of pK_w at various temperatures are listed in the table below.

Temperature (°C)	pK_w
0	14.9
10	14.5
20	14.2
30	13.8
40	13.5

Based on the information above, which of the following statements is true?

- (A) The dissociation of water is an exothermic process.
- (B) The pH of pure water is 7.00 at any temperature.
- (C) As the temperature increases, the pH of pure water increases.
- (D) As the temperature increases, the pH of pure water decreases. ✓

5.



A solution is prepared by mixing 50 mL of 1 M NaH_2PO_4 with 50 mL of 1 M Na_2HPO_4 . On the basis of the information above, which of the following species is present in the solution at the lowest concentration?

- (A) Na^+
- (B) HPO_4^{2-}
- (C) H_2PO_4^-
- (D) PO_4^{3-} ✓

6. Which of the following accounts for the observation that the pH of pure water at 37°C is 6.8?

- (A) At 37 °C water is naturally acidic.
- (B) At 37 °C the autoionization constant for water, K_w , is larger than it is at 25 °C. ✓
- (C) At 37 °C water has a lower density than it does at 25 °C; therefore, $[\text{H}^+]$ is greater.
- (D) At 37 °C water ionizes to a lesser extent than it does at 25 °C

Chapter 11 Part 1: Acidity



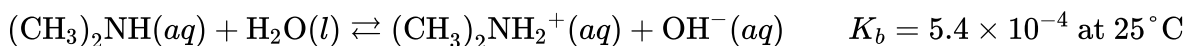
The equilibrium reaction shown above represents the partial ionization of the weak acid $\text{HCN}(aq)$. A 0.200 M $\text{HCN}(aq)$ solution has a $\text{pH} \approx 4.95$. If 0.05 g (0.010 mol) of $\text{NaCN}(s)$ is added to 100 mL of 0.200 M $\text{HCN}(aq)$, which of the following explains how and why the pH of the solution changes?

- (A) The pH will be higher than 4.95 because adding CN^- will disrupt the equilibrium, resulting in an increased production of HCN that decreases the concentration of H_3O^+ . ✓
- (B) The pH will be lower than 4.95 because adding CN^- will disrupt the equilibrium, resulting in an increased production of HCN that decreases the concentration of H_3O^+ .
- (C) The pH will be higher than 4.95 because CN^- is a strong base that can neutralize HCN .
- (D) The pH will remain close to 4.95 because the K_a is so small that hardly any products form.

8. At 25°C , aqueous solutions with a pH of 8 have a hydroxide ion concentration, $[\text{OH}^-]$, of

- (A) $1 \times 10^{-14}\text{ M}$
- (B) $1 \times 10^{-8}\text{ M}$
- (C) $1 \times 10^{-6}\text{ M}$ ✓
- (D) 1 M
- (E) 8 M

The equilibrium for the reaction between $(\text{CH}_3)_2\text{NH}$, a weak base, and water is represented by the equation below. The table shows the pH of three solutions of $(\text{CH}_3)_2\text{NH}(aq)$ at 25°C .



$[(\text{CH}_3)_2\text{NH}]$	pH at 25°C
0.050	11.69
0.10	11.85
0.20	12.01

9. Based on the information given, which of the following is true?

- (A) Solutions with a higher concentration of $(\text{CH}_3)_2\text{NH}$ have a higher pOH because to reach equilibrium a smaller amount of the conjugate acid $(\text{CH}_3)_2\text{NH}_2^+$ is produced.
- (B) Solutions with a higher concentration of $(\text{CH}_3)_2\text{NH}$ have a higher pOH because to reach equilibrium more OH^- is produced.
- (C) Solutions with a higher concentration of $(\text{CH}_3)_2\text{NH}$ have a higher pH because to reach equilibrium a smaller amount of the conjugate acid $(\text{CH}_3)_2\text{NH}_2^+$ is produced.
- (D) Solutions with a higher concentration of $(\text{CH}_3)_2\text{NH}$ have a higher pH because to reach equilibrium more OH^- is produced. ✓

Chapter 11 Part 1: Acidity

10. At 25 °C, enough distilled water is added to a 30.0 mL sample of $\text{HNO}_3(aq)$ with a pH of 4.20 so that the final pH of the diluted solution is 5.20. The volume of distilled water added to the original solution is closest to
- (A) 30.0 mL
(B) 60.0 mL
(C) 270. mL ✓
(D) 300. mL
11. How can 100. mL of sodium hydroxide solution with a pH of 13.00 be converted to a sodium hydroxide solution with a pH of 12.00 ?
- (A) By diluting the solution with distilled water to a total volume of 108 mL
(B) By diluting the solution with distilled water to a total volume of 200 mL
(C) By diluting the solution with distilled water to a total volume of 1.00 L ✓
(D) By adding 100. mL of 0.10 M HCl
(E) By adding 100. mL of 0.10 M NaOH
-

Refer to the following.

Concentration (M)	pH of Acid 1	pH of Acid 2	pH of Acid 3	pH of Acid 4
0.010	3.44	2.00	2.92	2.20
0.050	3.09	1.30	2.58	1.73
0.10	2.94	1.00	2.42	1.55
0.50	2.69	0.30	2.08	1.16
1.00	2.44	0.00	1.92	0.98

The pH of solutions of four acids prepared at various concentrations were measured and recorded in the table above.

The four acids are, in no particular order, chlorous, hydrochloric, lactic, and propanoic.

12. Of the following species, which has the greatest concentration in a 1.0 M solution of acid 1 at equilibrium?
- (A) OH^-
(B) H_3O^+
(C) Acid 1 ✓
(D) The conjugate base of acid 1
13. For which acid is the value of the acid-dissociation constant, K_a , the smallest?
-

Chapter 11 Part 1: Acidity

(A) Acid 1 ✓

(B) Acid 2

(C) Acid 3

(D) Acid 4

14. Which of the four acids listed in the table is hydrochloric acid?

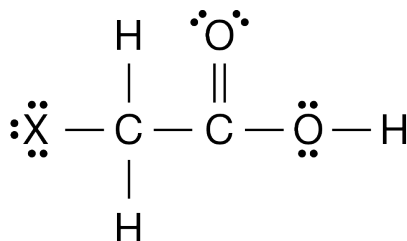
(A) Acid 1

(B) Acid 2 ✓

(C) Acid 3

(D) Acid 4

15. Samples of $\text{NaF}(s)$ and $\text{NH}_4\text{Cl}(s)$ are dissolved in separate beakers that each contain 100 mL of water. One of the salts produces a slightly acidic solution. Which of the following equations best represents the formation of the slightly acidic solution?

(A) $\text{Na}^+(aq) + 2\text{H}_2\text{O}(l) \rightleftharpoons \text{NaOH}(aq) + \text{H}_3\text{O}^+(aq)$ (B) $\text{F}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HF}(aq) + \text{OH}^-(aq)$ (C) $\text{NH}_4^+(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_3(aq) + \text{H}_3\text{O}^+(aq)$ ✓(D) $\text{Cl}^-(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{HCl}(aq) + \text{OH}^-(aq)$ 

The structure of haloacetic acids, XCH_2COOH (where X is either F, Cl, Br, or I), is shown above. The dissociation constants and molar masses of four haloacetic acids are listed in the table below.

Acid	$\text{p}K_a$	K_a	Molar Mass(g/mol)
Fluoroacetic acid	2.59	2.57×10^{-3}	78.0
Chloroacetic acid	2.87	1.35×10^{-3}	94.5
Bromoacetic acid	2.90	1.26×10^{-3}	138.9
Iodoacetic acid	3.18	6.61×10^{-4}	185.9

Chapter 11 Part 1: Acidity

16. An aqueous solution contains small but equal concentrations of both chloroacetic and fluoroacetic acids. Which statement comparing the percent ionizations of the two acids in the solution is true?

(A) The percent ionization of chloroacetic acid is greater than that of fluoroacetic acid.

(B) The percent ionization of chloroacetic acid is less than that of fluoroacetic acid. ✓

(C) The percent ionizations cannot be compared without knowing the concentrations of the two acids.

(D) The percent ionizations cannot be compared without knowing the pH of the solution.

17. A student mixes 100. mL of 0.25 M $\text{HCl}(aq)$ with 200. mL of 0.50 M $\text{HClO}_4(aq)$ and then dilutes the mixture with distilled water to a total volume of 500. mL. The $[\text{H}_3\text{O}^+]$ in the final solution is closest to

(A) 0.0025 M

(B) 0.12 M

(C) 0.25 M ✓

(D) 0.75 M

18. $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \rightleftharpoons \text{NH}_4^+(aq) + \text{OH}^-(aq)$ $K_b = 1.8 \times 10^{-5}$ at 25°C

Initial $[\text{NH}_3]$	Equilibrium $[\text{NH}_4^+]$	Equilibrium $[\text{OH}^-]$	pOH
0.15	1.6×10^{-3}	1.6×10^{-3}	2.78
0.30	2.3×10^{-3}	?	?

NH_3 is a weak base that reacts with water according to the chemical equilibrium represented above. The table provides some information for two $\text{NH}_3(aq)$ solutions of different concentration at 25°C . Which of the following is true about the more concentrated 0.30 M $\text{NH}_3(aq)$, and why?

(A) $[\text{OH}^-] = 3.2 \times 10^{-3} \text{ M}$ and $\text{pOH} < 2.78$, because a higher $[\text{OH}^-]$ corresponds to a lower pOH for 0.30 M $\text{NH}_3(aq)$ compared to 0.15 M $\text{NH}_3(aq)$.

(B) $[\text{OH}^-] = 3.2 \times 10^{-3} \text{ M}$ and $\text{pOH} > 2.78$, because a higher $[\text{OH}^-]$ corresponds to a higher pOH for 0.30 M $\text{NH}_3(aq)$ compared to 0.15 M $\text{NH}_3(aq)$.

(C) $[\text{OH}^-] = 2.3 \times 10^{-3} \text{ M}$ and $\text{pOH} < 2.78$, because a higher $[\text{OH}^-]$ corresponds to a lower pOH for 0.30 M $\text{NH}_3(aq)$ compared to 0.15 M $\text{NH}_3(aq)$. ✓

(D) $[\text{OH}^-] = 2.3 \times 10^{-3} \text{ M}$ and $\text{pOH} > 2.78$, because a higher $[\text{OH}^-]$ corresponds to a higher pOH for 0.30 M $\text{NH}_3(aq)$ compared to 0.15 M $\text{NH}_3(aq)$.

19. Is a strong electrolyte in aqueous solution

Chapter 11 Part 1: Acidity

- (A) $\text{NH}_3(\text{g})$
 (B) $\text{BH}_3(\text{g})$
 (C) $\text{H}_2(\text{g})$
 (D) $\text{H}_2\text{S}(\text{g})$

(E) $\text{HBr}(\text{g})$ ✓

20.

$\text{HOCl}(\text{aq})$	$K_a = 3.0 \times 10^{-8}$
$\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$	$K_a = 1.8 \times 10^{-5}$
$\text{NH}_3(\text{aq})$	$K_b = 1.8 \times 10^{-5}$
$\text{C}_6\text{H}_5\text{NH}_2(\text{aq})$	$K_b = 3.8 \times 10^{-10}$

The K_a or K_b values of some weak acids and bases are listed in the table above. Which of the following solutions has the highest pH?

- (A) $0.1\text{ M HOCl}(\text{aq})$
 (B) $0.1\text{ M HC}_2\text{H}_3\text{O}_2(\text{aq})$

(C) $0.1\text{ M NH}_3(\text{aq})$ ✓

(D) $0.1\text{ M C}_6\text{H}_5\text{NH}_2(\text{aq})$

21. The ionization constant for acetic acid is 1.8×10^{-5} ; that for hydrocyanic acid is 4×10^{-10} . In 0.1 M solutions of sodium acetate and sodium cyanide, it is true that

- (A) $[\text{H}^+]$ equals $[\text{OH}^-]$ in each solution
 (B) $[\text{H}^+]$ exceeds $[\text{OH}^-]$ in each solution
 (C) $[\text{H}^+]$ of the sodium acetate solution is less than that of the sodium cyanide solution

(D) $[\text{OH}^-]$ of the sodium acetate solution is less than that of the sodium cyanide solution ✓

(E) $[\text{OH}^-]$ for the two solutions is the same

22. Which of the following pairs of mathematical expressions can be used to correctly calculate the pH and pOH of a $0.0015\text{ M KOH}(\text{aq})$ solution at 25°C ?

- (A) $\text{pH} = -\log(14.00 - 0.0015)$ and $\text{pOH} = -\log(0.0015)$
 (B) $\text{pH} = \log(0.0015) - 14.00$ and $\text{pOH} = -\log(0.0015)$

(C) $\text{pH} = 14.00 - (-\log(0.0015))$ and $\text{pOH} = -\log(0.0015)$ ✓

(D) $\text{pH} = -(-\log(0.0015))$ and $\text{pOH} = 14.00 - (-\log(0.0015))$

23. Which of the following statements about the pH of 0.010 M HClO_4 is correct?

Chapter 11 Part 1: Acidity

(A) $\text{pH} = 2.00$, because $[\text{H}^+] = 1.0 \times 10^{-2} \text{ M}$. ✓

(B) $\text{pH} = 2.00$, because $[\text{H}^+] = 2.0 \times 10^{-2} \text{ M}$.

(C) $\text{pH} > 2.00$, because HClO_4 is a strong acid.

(D) $\text{pH} < 2.00$, because HClO_4 is a weak acid.

24. Which of the following is the correct mathematical expression to use to calculate the pH of a 0.10 M aqueous $\text{Ba}(\text{OH})_2$ solution at 25°C ?

(A) $\text{pH} = -\log(0.10)$

(B) $\text{pH} = -\log(0.20)$

(C) $\text{pH} = 14.00 - \log(0.10)$

(D) $\text{pH} = 14.00 + \log(0.20)$ ✓

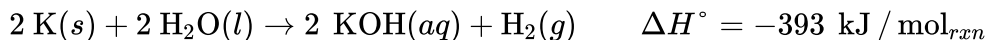
25. The value of K_w at 40°C is 3.0×10^{-14} . What is the pH of pure water at 40°C ?

(A) 3.0

(B) 6.8 ✓

(C) 7.0

(D) 7.2



When 3.9 g of $\text{K}(s)$ is added to $200. \text{ mL}$ of water at 25°C in a calorimeter, all the $\text{K}(s)$ reacts with the water as represented by the equation above.

26. Which of the following is true about the pH of the water after the reaction is complete?

(A) $\text{pH} < 7$

(B) $\text{pH} = 7$

(C) $\text{pH} > 7$ ✓

(D) The pH of the water cannot be determined without additional information.

27. In pure water, some of the molecules ionize according to the equation $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$. The extent of the ionization increases with temperature. A student heats pure water and records the measured pH at 50°C as 6.6 . Based on this information, which of the following mathematical relationships gives the pOH of pure water at 50°C ?

(A) $\text{pOH} = \text{pH}$ ✓

(B) $\text{pOH} = \frac{1}{\text{pH}}$

(C) $\text{pOH} = 14 - \text{pH}$

(D) $\text{pOH} = \frac{1 \times 10^{-14}}{\text{pH}}$

Chapter 11 Part 1: Acidity

28. The value of K_w for water at 0°C is 1×10^{-15} . What is the pOH of water at 0°C ?

(A) 6.5

(B) 7.0

(C) 7.5



(D) 8.0

29.

Acid	Concentration	pH
X	0.005 <i>M</i>	2.3
Y	2.0 <i>M</i>	2.8
Z	3.0 <i>M</i>	2.8

Which of the following correctly ranks the three monoprotic acids listed in the table above from the weakest to the strongest?

(A) $X < Y < Z$

(B) $X < Z < Y$

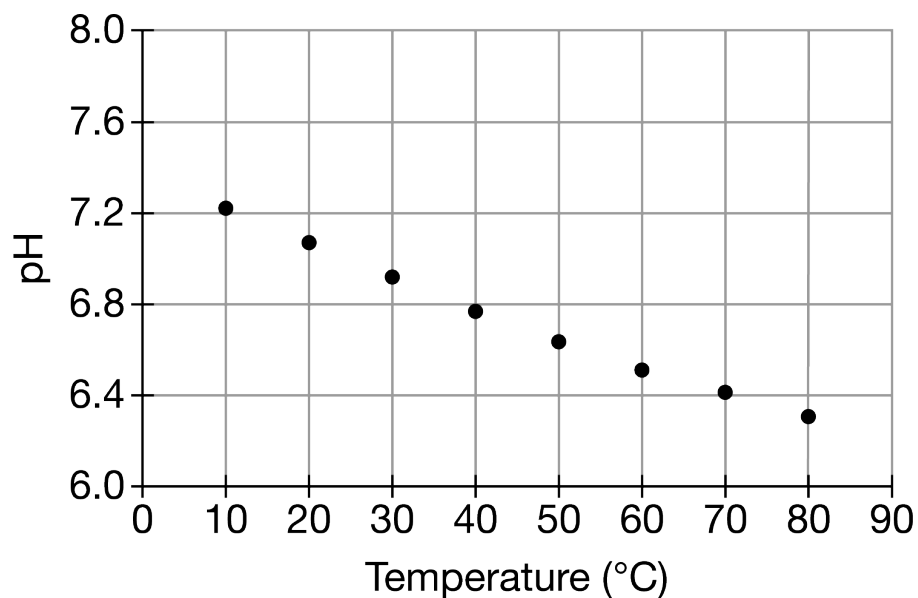
(C) $Y < Z < X$

(D) $Z < Y < X$



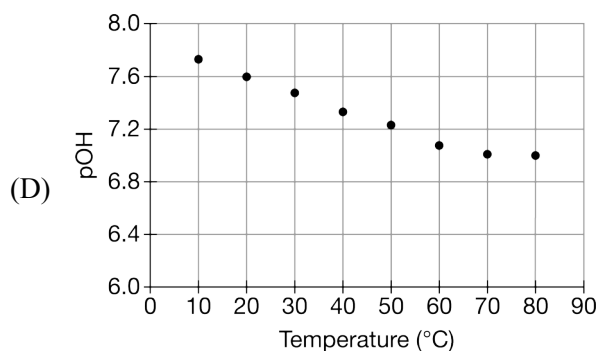
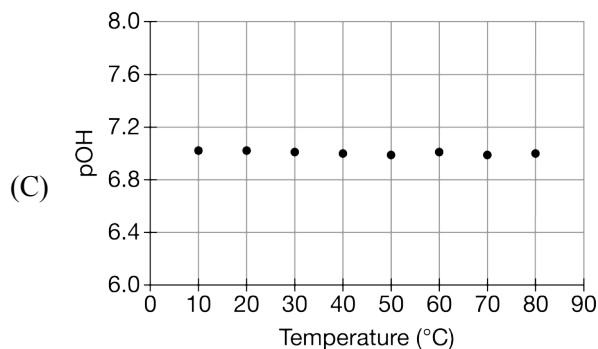
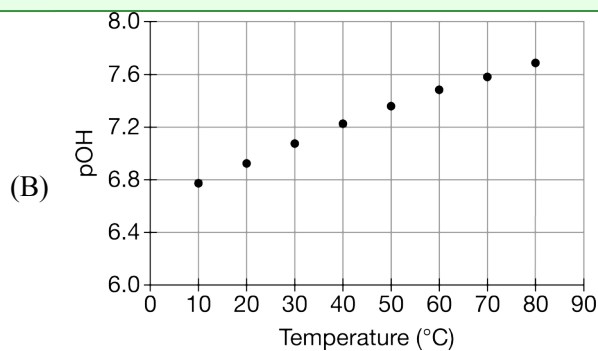
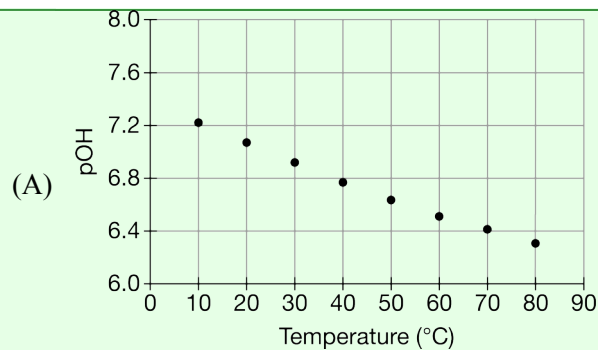
Chapter 11 Part 1: Acidity

30.

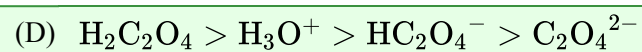
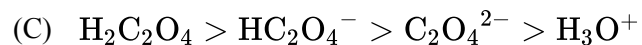
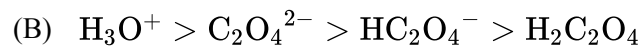
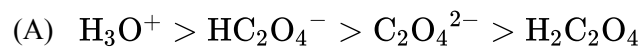


In pure water, some of the molecules ionize according to the equation $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$. The extent of the ionization increases with temperature. The graph above shows the pH of pure water at different temperatures. Which of the following represents the variations in the pOH of pure water under the same conditions?

Chapter 11 Part 1: Acidity



31. Which of the following identifies the four species that are present in an aqueous solution of the weak acid $\text{H}_2\text{C}_2\text{O}_4$, from the highest to the lowest concentration at equilibrium?



Chapter 11 Part 1: Acidity

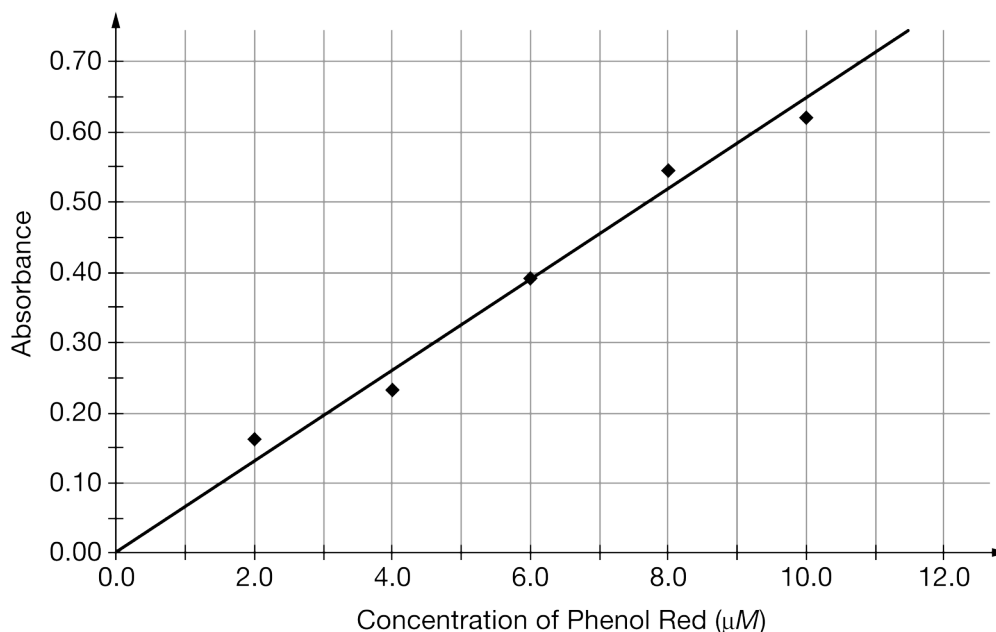
32. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student is given a solution of phenol red of unknown concentration. Solutions of phenol red are bright pink under basic conditions.

- (a) The student uses a pH probe and determines that the pH of the solution is 11.20. Calculate the hydrogen ion concentration, $[\text{H}^+]$, in the solution of phenol red.

The student analyzes a sample of the solution using a spectrophotometer set at a wavelength of 559 nm, the wavelength of maximum absorbance for phenol red. The measured absorbance of the phenol red solution at 559 nm is 0.35.

- (b) Based on the calibration curve shown below, what is the concentration of the solution in micromoles per liter (μM)?



- (c) If the student mixed 10 mL of distilled water with 10 mL of the sample, would this diluted solution have an absorbance greater than, less than, or equal to the absorbance of the original solution? Justify your answer.
- (d) If the student measured the absorbance of the solution at a wavelength of 650 nm, would the absorbance be greater than, less than, or equal to the absorbance of the solution at 559 nm? Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity

0	1
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The response gives a calculation equivalent to the following.

$$[\text{H}^+] = 10^{-\text{pH}} = 10^{-11.20} = 6.3 \times 10^{-12} \text{ M}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives a value between $5.2 \mu\text{M}$ and $5.6 \mu\text{M}$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ The response indicates that the diluted solution would have an absorbance less than that of the original solution.
- ☐ The response indicates that the diluted solution would have a lower concentration and therefore a lower absorbance, in accordance with Beer's law.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

Chapter 11 Part 1: Acidity

- ☐ The response indicates that the absorbance at 650 nm would be less than the absorbance at 559 nm.
- ☐ The response indicates that 559 nm is the wavelength of maximum absorbance so the absorbance at any other wavelength will be lower.

33. The pH of a solution prepared by the addition of 10. mL of 0.002 M KOH(aq) to 10. mL of distilled water is closest to

(A) 12

(B) 11

(C) 10

(D) 4

(E) 3

34. $\text{H}_2\text{O}(l) + \text{H}_2\text{O}(l) \rightleftharpoons \text{H}_3\text{O}^+(aq) + \text{OH}^-(aq)$

At 5.0°C, the value of K_w for the equilibrium shown above is 1.9×10^{-15} and the value of $\text{p}K_w$ is 14.73. Based on this information, which of the following is correct for pure water at this temperature?

(A) $[\text{H}_3\text{O}^+] = \sqrt{1.9 \times 10^{-15}}$

(B) $\text{pH} = -\log(1.9 \times 10^{-15})$

(C) $14.73 = [\text{H}_3\text{O}^+]_{eq}[\text{OH}^-]_{eq}$

(D) $\text{pOH} = \text{pH} + 14.73$

35. The value of K_w at 37°C is 2.4×10^{-14} . Which of the following statements about the pH and pOH of pure water at 37°C is correct?

(A) $\text{pH} < 7$ and $\text{pOH} > 7$ because water dissociates to a lesser extent at 37°C than at 25°C.

(B) $\text{pH} > 7$ and $\text{pOH} < 7$ because the dissociation of water is endothermic, and $\text{pH} + \text{pOH} = 14$.

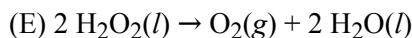
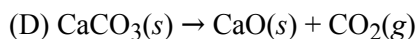
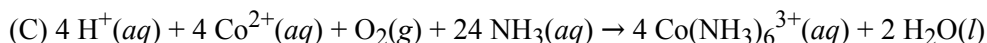
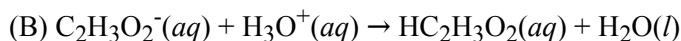
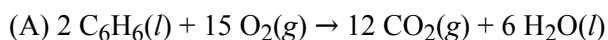
(C) $\text{pH} = \text{pOH} < 7$ because water dissociates to a greater extent at 37°C than at 25°C.

(D) $\text{pH} = \text{pOH} = 7$ because $[\text{H}_3\text{O}^+] = [\text{OH}^-]$ for pure water.

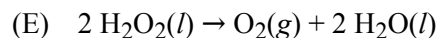
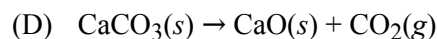
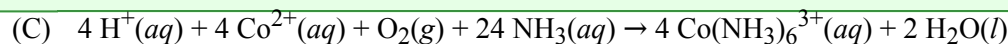
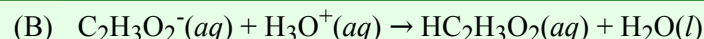
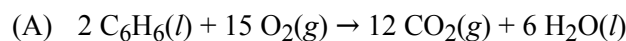
Chapter 11 Part 1: Acidity

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The following questions refer to the chemical reactions represented below.



36. The reaction between a Brønsted-Lowry acid and a Brønsted-Lowry base



37. Which of the following correctly identifies a Brønsted-Lowry conjugate acid-base pair in a solution prepared by mixing together $\text{HNO}_2(aq)$ and $\text{NH}_3(aq)$?

(A) The acid is $\text{HNO}_2(aq)$, and the conjugate base is $\text{NH}_3(aq)$.

(B) The acid is $\text{HNO}_2(aq)$, and the conjugate base is $\text{NO}_2^-(aq)$. ✓

(C) The acid is $\text{NH}_4^+(aq)$, and the conjugate base is $\text{NO}_2^-(aq)$.

(D) The acid is $\text{NH}_3(aq)$, and the conjugate base is $\text{NH}_4^+(aq)$.

38. Which of the following is the conjugate acid of NH_2^- ?

(A) NH^{2-}

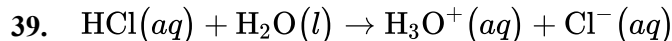
(B) NH_3 ✓

(C) H^+

(D) NH_4^+

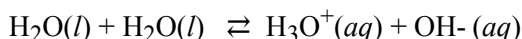
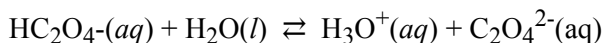
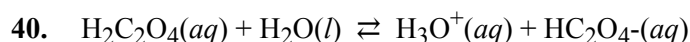
(E) H_2O

Chapter 11 Part 1: Acidity



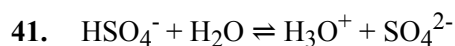
For the dissolution of HCl in water represented above, which of the following pairs includes the Brønsted-Lowry bases?

- (A) $\text{HCl}(aq)$ and $\text{Cl}^-(aq)$
(B) $\text{HCl}(aq)$ and $\text{H}_3\text{O}^+(aq)$
(C) $\text{H}_2\text{O}(l)$ and $\text{H}_3\text{O}^+(aq)$
(D) $\text{H}_2\text{O}(l)$ and $\text{Cl}^-(aq)$ ✓



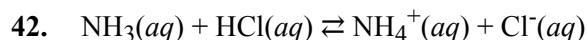
All the reactions represented above occur in an aqueous solution of oxalic acid. Which of the following represent a Brønsted-Lowry conjugate acid-base pair?

- (A) $\text{H}_2\text{C}_2\text{O}_4(aq)$ and $\text{C}_2\text{O}_4^{2-}(aq)$
(B) $\text{HC}_2\text{O}_4^-(aq)$ and $\text{C}_2\text{O}_4^{2-}(aq)$ ✓
(C) $\text{HC}_2\text{O}_4^-(aq)$ and $\text{H}_2\text{O}(aq)$
(D) $\text{H}_3\text{O}^+(aq)$ and $\text{OH}^-(aq)$



In the equilibrium represented above, the species that act as bases include which of the following?

- I. HSO_4^-
II. H_2O
III. SO_4^{2-}
(A) II only
(B) III only
(C) I and II
(D) I and III
(E) II and III ✓



The Brønsted-Lowry bases in the reaction represented above are

Chapter 11 Part 1: Acidity

(A) $\text{NH}_3(aq)$ and $\text{NH}_4^+(aq)$

(B) $\text{NH}_3(aq)$ and $\text{Cl}^-(aq)$



(C) $\text{NH}_3(aq)$ and $\text{HCl}(aq)$

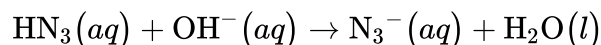
(D) $\text{HCl}(aq)$ and $\text{NH}_4^+(aq)$

(E) $\text{HCl}(aq)$ and $\text{Cl}^-(aq)$

Chapter 11 Part 1: Acidity

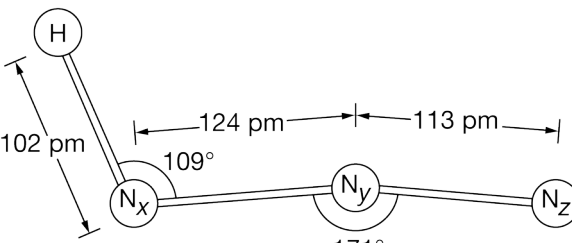
43. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Sodium azide, $\text{NaN}_3(s)$, is an ionic compound that can be synthesized from hydrazoic acid, $\text{HN}_3(aq)$, and $\text{NaOH}(aq)$, as represented by the net ionic equation below.



(a) The student believes this is an acid-base reaction and claims that HN_3 and OH^- are an acid-base conjugate pair. Do you agree or disagree with the claim that the two compounds are an acid-base conjugate pair? Justify your answer.

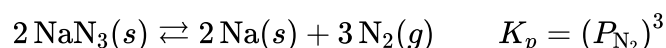
Two possible Lewis electron-dot diagrams and experimentally determined information for HN_3 are shown below.

Possible Lewis Electron-Dot Diagrams	
$\text{H}-\ddot{\text{N}}_{\text{x}}-\text{N}_{\text{y}}\equiv\text{N}_{\text{z}}:$ Diagram 1	$\text{H}-\ddot{\text{N}}_{\text{x}}=\text{N}_{\text{y}}=\ddot{\text{N}}_{\text{z}}$ Diagram 2
Experimentally Determined Information	
	

(b) Use VSEPR theory to explain why Diagram 1 is consistent with the experimentally determined bond angles around N_{y} shown above.

(c) Explain why Diagram 2 is NOT consistent with the experimentally determined bond lengths.

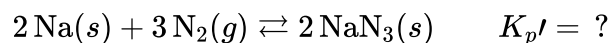
In a sealed, rigid and previously evacuated container, a solid sample of NaN_3 partially decomposes at 287 K, creating a volume of 75 L and an internal pressure of 1.1 atm. The decomposition reaction is represented by the equation below.



(d) Use the information above to calculate the value of K_p .

(e) The student finds a reference where the chemical equation is written differently, which is shown below. Calculate the value of K_p' for the reaction represented by the equation shown below.

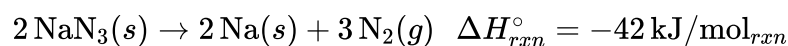
Chapter 11 Part 1: Acidity



(f) In a second experiment, the student runs the reaction under the same conditions using half of the initial mass of $\text{NaN}_3(s)$. After equilibrium is established, some $\text{NaN}_3(s)$ remains in the container. The student predicts the K_p value of the reaction will be cut in half in this experiment. Do you agree or disagree with the student's prediction? Justify your answer.

(g) In a third experiment, the student collects the nitrogen gas in a flexible container. The gas occupies a volume of 75 L at a pressure of 1.1 atm. If the pressure is changed to 0.55 atm with no change in temperature, determine the final volume. Include units in your answer.

The enthalpy of reaction when $\text{NaN}_3(s)$ decomposes at 298 K is shown below.



(h) A student claims that when 1.0 mol of NaN_3 is decomposed completely, 42 kJ of energy is absorbed by the reaction. Do you agree or disagree with the student's claim? Justify your answer in terms of both the sign and magnitude of $\Delta H_{\text{rxn}}^\circ$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The conjugate base of HN_3 is N_3^- , not OH^- . Acid-base conjugates differ only by one H^+ ion.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a correct explanation equivalent to the following.

· VSEPR theory predicts that the two bonding groups around N_y in diagram 1 will result in linear geometry with a bond angle close to 180° . This prediction is consistent with the experimentally determined bond angle of 171° .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity



0	1
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The response includes a correct explanation equivalent to the following.

· Diagram 2 implies that the $N_x - \text{to} - N_y$ and $N_y - \text{to} - N_z$ bonds will be equal in length because they are both $N = N$. The experimental data show that the $N_x - \text{to} - N_y$ bond is actually longer than the $N_y - \text{to} - N_z$ bond.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

$$\cdot K_p = (P_{N_2})^3 = (1.1)^3 = 1.3$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

$$\cdot K_p' = \frac{1}{(P_{N_2})^3} = \frac{1}{(1.1)^3} = 0.75$$

OR

$$\cdot K_p' = \frac{1}{K_p} = \frac{1}{1.3} = 0.77 \text{ (0.75 if not rounded from part (d))}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. Adding less $\text{NaN}_3(s)$ will not affect the partial pressure of $\text{N}_2(g)$, and therefore the value of K_p does not change. Only altering the temperature would change the value of K_p .

Part G

The response includes neither or only one of the following criteria.



0	1
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The response includes both of the following criteria.

- ☐ The response calculates the correct volume.
 $P_1V_1 = P_2V_2 \Rightarrow V_2 = \frac{(1.1 \text{ atm})(75 \text{ L})}{(0.55 \text{ atm})} = 150 \text{ L}$
 OR
 · The volume will be doubled when pressure is halved, so $75 \text{ L} \times 2 = 150 \text{ L}$.
- ☐ The response includes the correct units.
 · L

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a justification that addresses the sign AND magnitude of ΔH_{rxn}° equivalent to one of the following.

· Disagree. Since the reaction is exothermic ($\Delta H_{rxn}^\circ < 0$), energy is released by the system, not absorbed. Also, 42 kJ of energy is released when 2 moles of $\text{NaN}_3(s)$ reacts. Only 21 kJ of energy is released when 1 mole of $\text{NaN}_3(s)$ reacts.

OR

$$1 \text{ mol NaN}_3 \times \frac{1 \text{ mol rxn}}{2 \text{ mol NaN}_3} \times \frac{-42 \text{ kJ}}{1 \text{ mol rxn}} = -21 \text{ kJ}$$

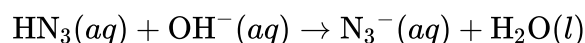
Chapter 11 Part 1: Acidity

Disagree. The magnitude of heat exchange is incorrect. Also, the reaction is exothermic ($\Delta H_{rxn}^{\circ} < 0$), which is inconsistent with the claim that heat is absorbed.

Chapter 11 Part 1: Acidity

44. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Sodium azide, $\text{NaN}_3(s)$, is an ionic compound that can be synthesized from hydrazoic acid, $\text{HN}_3(aq)$, and $\text{NaOH}(aq)$, as represented by the net ionic equation below.



(a) The student believes this is an acid-base reaction and claims that HN_3 and H_2O are an acid-base conjugate pair. Do you agree or disagree with the claim that the two compounds are an acid-base conjugate pair? Justify your answer.

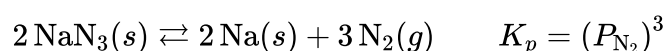
Two possible Lewis electron-dot diagrams and experimentally determined information for HN_3 are shown below.

Possible Lewis Electron-Dot Diagrams	
$\text{H}-\ddot{\text{N}}_x-\text{N}_y\equiv\text{N}_z:$ Diagram 1	$\text{H}-\ddot{\text{N}}_x=\text{N}_y-\ddot{\text{N}}_z:$ Diagram 2
Experimentally Determined Information	

(b) Use VSEPR theory to explain why Diagram 1 is consistent with the experimentally determined bond angles around N_x shown above.

(c) Explain why Diagram 2 is NOT consistent with the experimentally determined bond lengths.

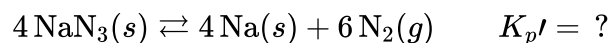
In a sealed, rigid and previously evacuated container, a solid sample of NaN_3 partially decomposes at 323 K, creating a volume of 75 L and an internal pressure of 1.2 atm. The decomposition reaction is represented by the equation below.



(d) Use the information above to calculate the value of K_p .

(e) The student finds a reference where the chemical equation is written differently, which is shown below. Calculate the value of K_p for the reaction represented by the equation shown below.

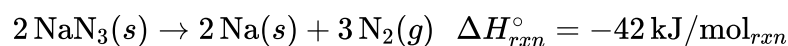
Chapter 11 Part 1: Acidity



(f) In a second experiment, the student runs the reaction under the same conditions using double the initial mass of $\text{NaN}_3(s)$, but the volume of $\text{NaN}_3(s)$ is insignificant compared to that of the gas. The student predicts the K_p value of the reaction will double in this experiment. Do you agree or disagree with the student's prediction? Justify your answer.

(g) In a third experiment, the student collects the nitrogen gas in a flexible container. The gas occupies a volume of 75 L at a pressure of 1.2 atm. If the pressure is changed to 2.4 atm with no change in temperature, determine the final volume. Include units in your answer.

The enthalpy of reaction when $\text{NaN}_3(s)$ decomposes at 298 K is shown below.



(h) A student claims that when 1.0 mol of NaN_3 is decomposed completely, 21 kJ of energy is released into the surroundings. Do you agree or disagree with the student's claim? Justify your answer in terms of both the sign and magnitude of $\Delta H_{\text{rxn}}^\circ$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The conjugate base of HN_3 is N_3^- , not H_2O . Acid-base conjugates differ only by one H^+ ion.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a correct explanation equivalent to the following.

· VSEPR theory predicts that the two bonding pairs and two lone pairs around N_x in diagram 1 will result in a bent geometry with a bond angle close to 109.5° . This prediction is consistent with the experimentally determined bond angle of 109° .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity

0	1
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The response includes a correct explanation equivalent to the following.

· Diagram 2 implies that the $N_x - \text{to} - N_y$ bond will be shorter than the $N_y - \text{to} - N_z$ bond due to the double bond between N_x and N_y and the single bond between N_y and N_z . The experimental data shows the opposite to be true of the two bond lengths.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

· $K_p = (P_{N_2})^3 = (1.2)^3 = 1.7$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

· $K_p' = (P_{N_2})^6 = (1.2)^6 = 3.0$

OR

· $K_p' = (K_p)^2 = (1.7)^2 = 2.9$ (3.0 if not rounded from part (d))

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. Adding $\text{NaN}_3(s)$ will not affect the partial pressure of $\text{N}_2(g)$, and therefore the value of K_p does not change. Only altering the temperature would change the value of K_p .

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both of the following criteria.

- ☐ The response calculates the correct volume.
 $P_1V_1 = P_2V_2 \Rightarrow V_2 = \frac{(1.2 \text{ atm})(75 \text{ L})}{(2.4 \text{ atm})} = 38 \text{ L}$
 OR
 · The volume will be halved when the pressure is doubled, so $\frac{75 \text{ L}}{2} = 38 \text{ L}$.
- ☐ The response includes the correct units.
 · L

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response does not provide both the correct answer and a justification that addresses the sign AND magnitude of ΔH_{rxn}° equivalent to one of the following.

· Agree. Since the reaction is exothermic ($\Delta H_{rxn}^\circ < 0$), energy is released by the system. There is 42 kJ of energy released for every 2 moles of $\text{NaN}_3(s)$ that reacts, or 21 kJ for every 1 mole of $\text{NaN}_3(s)$ that reacts.

OR

· $1 \text{ mol NaN}_3 \times \frac{1 \text{ mol rxn}}{2 \text{ mol NaN}_3} \times \frac{-42 \text{ kJ}}{1 \text{ mol rxn}} = -21 \text{ kJ}$

Agree. The reaction is exothermic ($\Delta H_{rxn}^\circ < 0$) and releases 21 kJ of energy when 1 mol of $\text{NaN}_3(s)$ reacts

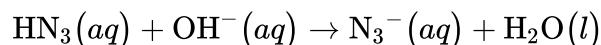
Chapter 11 Part 1: Acidity

completely.

Chapter 11 Part 1: Acidity

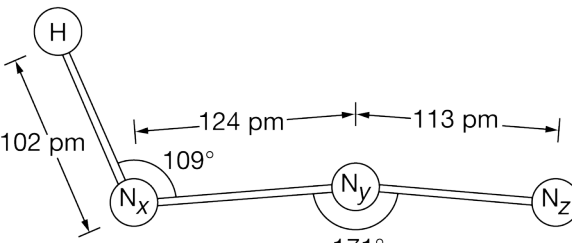
45. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Sodium azide, $\text{NaN}_3(s)$, is an ionic compound that can be synthesized from hydrazoic acid, $\text{HN}_3(aq)$, and $\text{NaOH}(aq)$, as represented by the net ionic equation below.



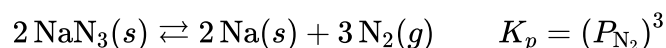
(a) The student believes this is an acid-base reaction and claims that HN_3 and N_3^- are an acid-base conjugate pair. Do you agree or disagree with the claim that the two chemical species are an acid-base conjugate pair? Justify your answer.

Two possible Lewis electron-dot diagrams and experimentally determined information for HN_3 are shown below.

Possible Lewis Electron-Dot Diagrams	
$\text{H}-\ddot{\text{N}}_x-\ddot{\text{N}}_y=\ddot{\text{N}}_z$ Diagram 1	$\text{H}-\ddot{\text{N}}_x-\text{N}_y\equiv\text{N}_z:$ Diagram 2
Experimentally Determined Information	
	

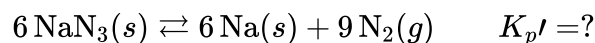
- (b) Explain why Diagram 2 is consistent with the experimentally determined bond lengths shown above.
- (c) Use VSEPR theory to explain why Diagram 1 is NOT consistent with the experimentally determined bond angles around N_y .

In a sealed, rigid and previously evacuated container, a solid sample of NaN_3 partially decomposes at 340. K creating a volume of 75 L and an internal pressure of 1.3 atm. The decomposition reaction is represented by the equation below.



- (d) Use the information above to calculate the value of K_p .
- (e) The student finds a reference where the chemical equation is written differently, which is shown below. Calculate the value of K_p' for the reaction represented by the equation shown below.

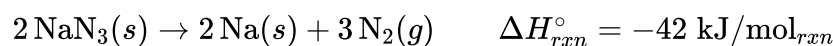
Chapter 11 Part 1: Acidity



(f) In a second experiment, the student runs the reaction under the same conditions using a smaller 35 L container, but the volume of $\text{NaN}_3(s)$ is insignificant compared to that of the gas. The student predicts the K_p value of the reaction will decrease in this experiment. Do you agree or disagree with the student's prediction? Justify your answer.

(g) In a third experiment, the student collects the nitrogen gas in a separate rigid container. The gas occupies a volume of 75 L at 1.3 atm and 340 K. If the temperature is changed to 170 K at constant volume, determine the final pressure. Include units in your answer.

The enthalpy of reaction when $\text{NaN}_3(s)$ decomposes at 298 K is shown below.



(h) A student claims that when 6.0 mol of NaN_3 is decomposed completely, 126 kJ of energy is absorbed by the reaction. Do you agree or disagree with the student's claim? Justify your answer in terms of both the sign and magnitude of $\Delta H_{\text{rxn}}^\circ$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Agree. When HN_3 donates a proton, it forms the conjugate base, N_3^- . Acid-base conjugates differ only by one H^+ ion.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes a correct explanation equivalent to the following.

· Diagram 2 has a single bond between N_x and N_y and a triple bond between N_y and N_z . Triple bonds are shorter than single bonds, so these predictions about relative bond length are consistent with the experimental data.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity

0	1
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The response does not include a correct explanation equivalent to the following.

· VSEPR theory predicts that the two bonding groups and one lone pair around N_y in diagram 1 will result in a bent geometry with a bond angle close to 120° . This prediction is inconsistent with the experimentally determined bond angle of 171° .

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

$$\cdot K_p = (P_{N_2})^3 = (1.3)^3 = 2.2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer.

$$\cdot K_p' = (P_{N_2})^9 = (1.3)^9 = 11$$

OR

$$\cdot K_p' = (2.2)^3 = 11$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 11 Part 1: Acidity



0	1
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The response indicates disagreement and provides a valid justification equivalent to the following.

· Disagree. If the volume is decreased, more $\text{NaN}_3(s)$ will re-form to re-establish the original pressure, and hence K_p does not change. Only altering the temperature would change the value of K_p .

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both of the following criteria.

- ☐ The response calculates the correct pressure.
 $\cdot \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow P_2 = \frac{(1.3 \text{ atm})(170 \text{ K})}{(340 \text{ K})} = 0.65 \text{ atm}$
 OR
 $\cdot$ The pressure will be halved when the temperature is halved, so $\frac{1.3 \text{ atm}}{2} = 0.65 \text{ atm}$.
- ☐ The response includes the correct units.
 $\cdot \text{atm}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a justification that addresses the sign AND magnitude of ΔH_{rxn}° equivalent to one of the following.

· Disagree. It is true that 126 kJ of energy is exchanged when 6 moles of $\text{NaN}_3(s)$ reacts, but energy is released, not absorbed, in this process because the reaction is exothermic ($\Delta H_{rxn}^\circ < 0$).

OR

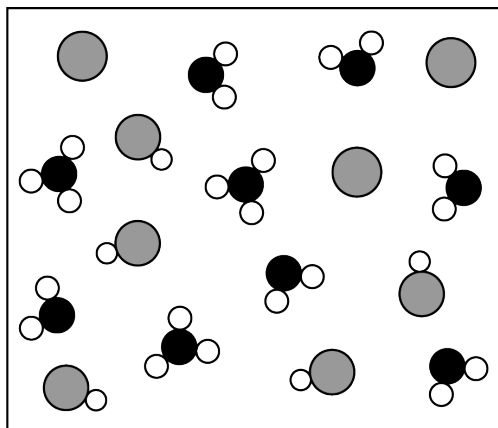
· $6 \text{ mol NaN}_3 \times \frac{1 \text{ mol rxn}}{2 \text{ mol NaN}_3} \times \frac{-42 \text{ kJ}}{1 \text{ mol rxn}} = -126 \text{ kJ}$

Disagree. The magnitude of heat exchange is correct, but the reaction is exothermic ($\Delta H_{rxn}^\circ < 0$), which is inconsistent

Chapter 11 Part 1: Acidity

with the claim that heat is absorbed.

46.



The particle diagram above represents an aqueous solution of a weak monoprotic acid. The white circles represent H atoms. Which of the following shows the species that act as a Brønsted-Lowry base and its conjugate acid, in that order, in the solution?

